

The Soficity Trace

What was tried, what failed, what survived, and what it cost

Jeromie Beasley

23 August 2026

Abstract

A complete working record of one attempt to bring the operator-first winding currency to bear on the existence of non-sofic groups. It contains the failed gate at full size, the theorem hidden inside the failure, the repaired gate and its eight control blocks, the exact conversions between the four metrics that the question lives on, the three results that went into the projector geometry paper — integrality and invariance of the winding, the admissible domain $\delta + g^2 \leq 4$, and the amplitude floor $|w| \leq (d/4)\sqrt{\delta}$ — the standing verdict that the route does not reach soficity and exactly why, and the routes killed and left open. Every claim carries a grade: [T] derived, [V] gated numerically, [C] cited, [H] interpretation. Nothing that was killed has been deleted.

1 The situation

On 1 August 2026 OpenAI announced ten machine-verified results from an unreleased model, Astra. The third is the construction of a non-sofic group, closing a question posed by Gromov in 1999 and named by Weiss in 2000 [C]. A group is sofic if it can be approximated by finite permutations: for any finite set of elements and any ε , there is a map into a symmetric group that is almost-multiplicative (composition agrees on at least $1 - \varepsilon$ of the points) and almost-free (any non-identity element moves at least $1 - \varepsilon$ of them). The metric is normalised Hamming — the fraction of points on which two permutations differ.

Audit of the released repository, run 20 August: all ten formalisation files carry zero `sorry`, zero `axiom` declarations, and zero `native_decide` — the last matters, because it is the standard way a zero-sorry proof still escapes the kernel. The forty-two sorries in the repository are confined to a separate challenge-stub directory and are not part of the proofs. This is a claim about a repository at a date, not a mathematical theorem, and it is recorded as one [V for the mechanical audit]. What a certificate cannot settle is whether the formalised statement is the intended one; that surface is unaudited [H].

Follow-up, checked directly:

- Fournier-Facio, arXiv:2608.02025 (3 August). Takes the key technical criterion and produces a finitely presented torsion-free non-sofic group, avoiding Leavitt algebras and Thompson’s V entirely, by small cancellation over a relatively hyperbolic pair. The stated point of the note is that the criterion, not the construction, is the novelty [C].
- Alekseev–Thom, arXiv:2608.05362. The centraliser of a sofic embedding of a Kazhdan group is itself a metric ultraproduct of permutation groups; an ergodic centraliser forces LEF, and finite presentation then forces residual finiteness [C].

- Kun–Thom, arXiv:2608.06222. Generalises the mechanism: for $\Gamma < G$ with the commensurator of Γ generating G , both Kazhdan, Γ not normal, the generalised wreath product $(\bigoplus_{G/\Gamma} \mathbb{Z}/2) \rtimes G$ is non-sofic — on explicit pairs of residually finite Kazhdan elementary matrix groups over polynomial rings. Also separates weakly sofic from sofic [C].

As of the date of this trace nothing further has appeared, and the two siblings, MF and hyperlinear, remain open [C]. Both halves of that sentence are dated on purpose.

2 The map onto the framework, before any verdict

Stated first because the comparison is worth making in earnest, and because several of the correspondences are exact rather than suggestive. Where a correspondence is exact it is derived; where it is a translation into this framework’s vocabulary it is labelled as one.

2.1 The two clauses

The two sofic clauses are the framework’s two. Almost-multiplicative is cocycle composition. For almost-free the sofic condition is a Hamming condition: a non-identity element’s approximating permutation must move almost every point. On the framework side the analogue is a uniform principal-angle separation, $\sin^2 \theta \geq \varepsilon$ — distinct directions stay resolvable [H]. Compose correctly, and do not lose anybody. The translation is the right one, and it is a translation, not a definition of soficity.

2.2 Four metrics, and the exact rates between them

The question lives on a ladder of metrics that are routinely conflated, and the whole verdict of this trace turns on the difference between the rungs. Fix $d \times d$ matrices and write

$$\|T\|_{\text{op}} = \sup_{\|x\|=1} \|Tx\|, \quad \|T\|_F = \left(\sum_{ij} |T_{ij}|^2 \right)^{1/2}, \quad \|T\|_{\text{HS},n} = \frac{\|T\|_F}{\sqrt{d}},$$

and for permutations σ, τ of d points the normalised Hamming distance $d_H(\sigma, \tau) = \frac{1}{d} \#\{i : \sigma(i) \neq \tau(i)\}$.

The two conversions that matter are exact. For permutation matrices P, Q the i -th row of $P - Q$ is either zero or a difference of two distinct standard basis vectors, so

$$\|P - Q\|_F^2 = 2 \#\{i : P(i) \neq Q(i)\}, \quad \text{hence} \quad \boxed{d_H(P, Q) = \frac{1}{2} \|P - Q\|_{\text{HS},n}^2}$$

[T]. Normalised Hamming is therefore not normalised Hilbert–Schmidt; it is half its square. Applied to the closure defect of this trace, $\delta = \|R - I\|_F^2/d$, that gives

$$\boxed{\delta = 2 d_H(R, I)} \quad \text{for } R \text{ a permutation matrix,}$$

which is the bridge from the gate’s amplitude to the sofic metric, with the factor written down [T]. Nothing measured in this trace is a permutation matrix — the shift is one, the clock is not — so the identity is never used to convert a reported number. It is here because the verdict of Section 5 is a statement about what does and does not cross between these metrics, and that statement needs the rate between them stated exactly rather than gestured at.

The projector side carries the same warning about normalisation. For equal-rank orthogonal projectors, $\|\Pi_1 - \Pi_2\|_F^2 = 2 \sum_i \sin^2 \theta_i$; that is a sum, not a fraction. The fraction of the subspace on which they disagree is the normalised quantity

$$\frac{1}{2r} \|\Pi_1 - \Pi_2\|_F^2 = \frac{1}{r} \sum_i \sin^2 \theta_i,$$

r the common rank [T]. The framework's observables are means of $\sin^2$ of principal angles, and it is the mean, not the sum, that makes ambient size drop out — which is the content of the zero-and-span lemma.

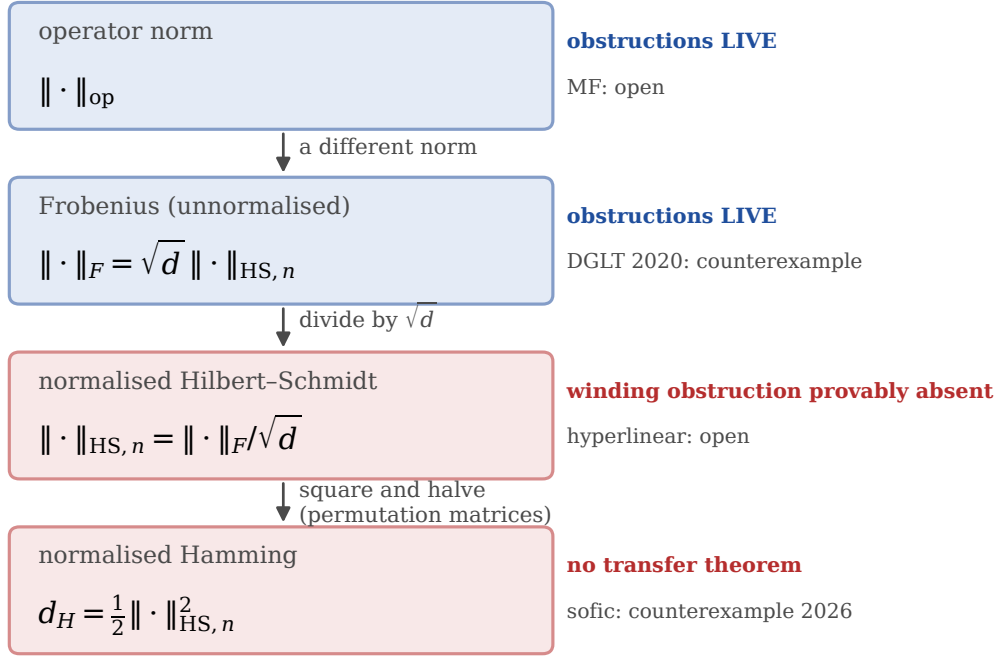


Figure 1: The four metrics, the exact rates between them, and what survives each. Descending the ladder is not a change of units: the normalisation is the whole content, and obstructions that live on the upper two rungs are not available on the lower two.

These four are not interchangeable, and the ladder is the reason a group can fail approximation in one sense while nothing is known in another. Concretely: change a permutation on εd points. Normalised Hamming distance ε , vanishing. Operator norm distance 2, the maximum.

2.3 The DGLT door, and which rung it is on

De Chiffre–Glebsky–Lubotzky–Thom (2017; published 2020) built the first non-approximable groups from central extensions $1 \rightarrow \mathbb{Z} \rightarrow \tilde{G} \rightarrow G \rightarrow 1$ with the obstruction a class in $H^2(G; \mathbb{Z})$. Their norm is the Frobenius norm, $\|T\|_F = (\sum_{ij} |T_{ij}|^2)^{1/2}$ — the *unnormalised* Hilbert–Schmidt norm, $\sqrt{d}$ times the normalised one [C]. That is not this framework's metric natively. Ambient size does not drop out of it, which is exactly the opposite of the zero-and-span normalisation, and it is the reason a counterexample exists on that rung while the hyperlinear question — the same norm, normalised — has been open since Connes.

The structural rhyme survives the correction, and is worth keeping for what it is. A central extension is a phase; the obstruction class is a winding no approximation can unwind [H]. What the correction removes is the claim that the door was already open onto this framework's own floor. It was one rung up.

2.4 Property (T) is a spectral gap

The trivial representation is isolated; there are no almost-invariant vectors. Its role in killing soficity is a rigidity argument: approximation is forced near something exact which cannot be there. That is a wall in the framework's sense [H].

2.5 The honest blocker, stated after the map

The framework's instruments live in finite dimensions with isolated spectrum: Gram determinants, principal angles, contours at $\eta \gtrsim \sqrt{\varepsilon}$. Sofic approximations live in an ultraproduct of symmetric groups where the spectrum is a measure and there is nowhere to put a contour. Second, and structurally worse: the diagnostic is always given an operator family and asks whether it tracks. Soficity asks whether *any* approximating family exists. That is a quantifier flip, and persistence of a transport one has says nothing about non-existence of transports one does not.

3 WIND-GATE-01 — built, run, failed

3.1 The design

One closure defect, two readings. Amplitude $\delta = \|R - I\|_F^2/d$. Phase $w = (1/2\pi)$ times the unwrapped total change in $(1/d) \arg \det R^m$ over $m = 0, \dots, d$. Testbed: clock and shift in dimension d , relator $R = ABA^{-1}B^{-1} = \omega^{-1}I$, a genuine 2-cocycle, with $\delta = 4 \sin^2(\pi/d) \rightarrow 0$.

3.2 What passed, and it is the half worth keeping

P1, exact control: PASS. $\delta = 1.9 \times 10^{-32}$, $w = 0$.

P2, noise control: PASS. δ from 2.4×10^{-1} down to 2.3×10^{-7} across $\eta = 10^{-1} \dots 10^{-4}$, with $w = 0.000000$ at every level. The null side is clean: the gate does not manufacture obstructions out of small amplitude defects.

3.3 What failed

P3: FAIL. δ reproduced exactly — $4 \sin^2(\pi/d)$ to every digit, a $715\times$ drop from $d = 3$ to $d = 97$ — but $w = 0$ at every d .

P4: FAIL. $w(cR) \neq w(R)$. Since $\det(cR^m) = c^m \det(R^m)$, the phase shift is m -dependent and does not cancel. Fails zero-and-span outright.

P5: FAIL. A matched- δ pair returned $w = 0.103322$ — not an integer. A winding needs a closed path; R^m for $m = 0, \dots, d$ closes only when $R^d = I$. The unperturbed clock-shift closes, the perturbed one does not, so $\arg \det$ simply accumulates as $m\theta$. An integer was asserted and closure was never gated.

3.4 The special-case manifestation

$\det(\zeta I_d) = \zeta^d$, and by Schur an irreducible projective representation whose commutator has order p has dimension exactly p , so $\zeta^d = 1$ always. In the dimension where the projective representation lives, the determinant line is structurally blind to a scalar 2-cocycle [T]. No finite-dimensional projective representation has a cocycle that $\det$ can see. This is exactly why DGLT needed an approximation norm and a cohomological argument rather than a determinant: everyone who tries the obvious phase invariant hits this wall. The risk was flagged during design and then talked past; it should have been gate one, not gate three.

3.5 The general diagnosis

The blindness above is true and it is not the root cause. The root cause is the path. The determinant of a commutator is identically 1, so the endpoint of any commutator carries exactly zero information, and no adjustment of dimension repairs that. The correct loop is Kazhdan's, rediscovered by Exel and Loring in 1991: the chord $t \mapsto \det((1-t)I + tR)$, $t \in [0, 1]$, which closes precisely because $\det = 1$ at both ends. Its winding number equals the summed principal arguments of the eigenvalues of R — the quantity physicists now call the Bott index [C].

The relation between the two readings is one line: $\det = \exp \operatorname{Tr} \log$, so the determinant collapses the summed principal logarithms modulo $2\pi i$. It discards exactly one wrap, and the wrap is the entire invariant.

4 WIND-GATE-02 — repaired, run, passed

4.1 Why the check proves what it claims, stated before the result

For clock and shift, $R = \omega I$ with $\omega = e^{2\pi i/d}$. Summing the principal logarithms of the d eigenvalues gives $d \cdot (2\pi i/d) = 2\pi i$, so $w = 1$. Taking the determinant instead exponentiates that sum and collapses it to $\omega^d = 1$. The two differ by one wrap. The check therefore has a known exact answer at every d , which makes the run a calibration against a fixed point rather than a discovery: anything other than 1 on the first block means the instrument is broken, not the mathematics. What makes it worth running at all is that $\delta = 4 \sin^2(\pi/d) \rightarrow 0$ while w stays pinned at 1 — an amplitude dying while an integer persists, with an answer known in advance to check against.

4.2 The results

Theorem 1 (Integrality and invariance; T+V). *Let $g = \min_j |\lambda_j + 1| > 0$, where λ_j are the eigenvalues of $R = UVU^\dagger V^\dagger$, and let $w = (1/2\pi) \sum_j \arg \lambda_j$ on the principal branch. Then $w \in \mathbb{Z}$ with $|w| \leq d/2$; w is unchanged by simultaneous unitary conjugation; w is unchanged by $U \mapsto e^{i\theta} U$ and $V \mapsto e^{i\varphi} V$; and w equals the winding number of $t \mapsto \det((1-t)I + tR)$.*

Proof. R is a commutator of unitaries, so $\det R = 1$, hence $\prod_j \lambda_j = 1$ and $\sum_j \arg \lambda_j \in 2\pi\mathbb{Z}$. The bound follows from $|\arg \lambda_j| < \pi$. Conjugation preserves the spectrum. The scalars cancel identically inside the commutator. The chord closes because $\det R = 1$ at $t = 1$ and $\det I = 1$ at $t = 0$; the identification of its winding with the summed principal arguments is Exel–Loring [C]. $\square$

The determinant of the relator is identically one. That triviality is not an obstacle to reading the record; it is the reason the record is an integer.

Remark 2 (The sharp integer bound). When the guard is active the bound $|w| \leq d/2$ is never attained. Every principal argument satisfies $|\arg \lambda_j| < \pi$ strictly, so $|w| < d/2$, and since w is an integer,

$$|w| \leq \left\lfloor \frac{d-1}{2} \right\rfloor$$

[T]. At $d = 31$ this reads $|w| \leq 15$, and the ladder of Proposition 8 attains it.

Theorem 3 (The admissible domain; T+V). *For every $U, V \in U(d)$, with $\delta = \|R - I\|_F^2/d$,*

$$\delta + g^2 \leq 4,$$

with equality if and only if $\operatorname{Re} \lambda_j$ is constant over the spectrum of R .

Proof. Each λ_j is on the unit circle, so $|\lambda_j - 1|^2 + |\lambda_j + 1|^2 = 2|\lambda_j|^2 + 2 = 4$. Averaging over j gives $\delta + d^{-1} \sum_j |\lambda_j + 1|^2 = 4$ exactly. Since g^2 is the minimum of the quantities being averaged, $g^2 \leq 4 - \delta$. Equality holds when every $|\lambda_j + 1|^2 = 2 + 2 \operatorname{Re} \lambda_j$ takes the same value. $\square$

Corollary 4 (Forced refusal). *The branch guard is capped by the defect. No anchor with $\delta > 4 - \tau^2$ is admissible at tolerance τ , in any dimension, for any pair of generators. A large defect and a safe branch cannot coexist.*

The half-angle form is what puts this in the corpus's own currency. Write $\lambda_j = e^{i\vartheta_j}$. Then $|\lambda_j - 1|^2 = 4 \sin^2(\vartheta_j/2)$ and $|\lambda_j + 1|^2 = 4 \cos^2(\vartheta_j/2)$, so the defect is four times the mean of $\sin^2$ of the half-angles and the guard squared is four times their minimum $\cos^2$. The bound reads

$$\operatorname{mean}_j \sin^2(\vartheta_j/2) + \min_j \cos^2(\vartheta_j/2) \leq 1 :$$

the Pythagorean identity with a minimum standing where a mean belongs [T]. The projector observables of the framework are natively $\sin^2$ -valued in principal angles; the defect is a squared sine, the guard is its cosine, and the wall is the right angle.

Theorem 5 (Amplitude floor; T+V). *For every U, V with $g > 0$,*

$$|w| \leq \frac{d}{4} \sqrt{\delta}, \quad \text{hence} \quad w \neq 0 \implies \delta \geq \frac{16}{d^2}.$$

Proof. $|\lambda_j - 1| = 2|\sin(\vartheta_j/2)|$, and $\sin x \geq 2x/\pi$ on $[0, \pi/2]$, so $|\vartheta_j| \leq \frac{\pi}{2} |\lambda_j - 1|$. Summing and applying Cauchy–Schwarz, $2\pi|w| \leq \sum_j |\vartheta_j| \leq \frac{\pi}{2} \sum_j |\lambda_j - 1| \leq \frac{\pi}{2} \sqrt{d \sum_j |\lambda_j - 1|^2} = \frac{\pi}{2} d \sqrt{\delta}$. $\square$

Corollary 6 (Higher winding). *Squaring, for any nonzero integer w ,*

$$\delta \geq \frac{16 w^2}{d^2}.$$

The floor is quadratic in the winding, not merely positive: carrying a larger integer costs quadratically more amplitude at fixed dimension [T].

An integer record cannot ride on an arbitrarily small defect. The amplitude may collapse as dimension grows — and on clock and shift it does — but not faster than d^{-2} , and a record of size w not faster than $w^2 d^{-2}$.

The ancestry of Theorem 5 should be stated plainly. Quantitative lower bounds on how far almost-commuting unitaries must move before a Bott-type index can change are Loring's, in the operator norm and dimension-independently (arXiv:1302.0349) [C]. What is contributed here is the normalised-Frobenius form — necessarily dimension-dependent, since the normalisation is what makes it so — together with the clock–shift calibration below, which shows the constant cannot be improved.

Proposition 7 (Clock and shift). *With $U = \text{diag}(1, \omega, \dots, \omega^{d-1})$ and V the cyclic shift, $UV = \omega VU$, $R = \omega I$, and $\delta = 4 \sin^2(\pi/d)$, $g = 2 \cos(\pi/d)$, $w = 1$ for $d \geq 3$. Every half-angle is π/d , so the family realises equality in the domain bound at every d and saturates the floor up to the constant $\pi/2$. At $d = 2$ the half-angle reaches $\pi/2$, $R = -I$, $\delta = 4$, $g = 0$, and the anchor is inadmissible.*

Proposition 8 (The winding ladder; T+V). *Fix d and replace U by U^k . Then $R = \omega^k I$ and, for $1 \leq k \leq \lfloor (d-1)/2 \rfloor$,*

$$w = k, \quad \delta_k = 4 \sin^2\left(\frac{k\pi}{d}\right), \quad g_k = 2 \cos\left(\frac{k\pi}{d}\right), \quad \delta_k + g_k^2 = 4.$$

The whole ladder lies on the equality locus of the domain bound. Against Corollary 6 it gives

$$\frac{\delta_k}{16k^2/d^2} = \left(\frac{d}{2k} \sin \frac{k\pi}{d}\right)^2,$$

which decreases monotonically from $\pi^2/4$ as $k/d \rightarrow 0$ to 1 as $k \rightarrow d/2$.

Corollary 9 (The constant is optimal). *The factor $1/4$ in Theorem 5 cannot be improved. Along the ladder, $(d/4)\sqrt{\delta_k}/|w| = (d/2k) \sin(k\pi/d) \rightarrow 1$ as $k \rightarrow d/2$, so the inequality is asymptotically an equality on an explicit admissible family [T]. The loss factor $\pi/2$ visible on the $w = 1$ family is a small-angle artefact, not slack in the theorem.*

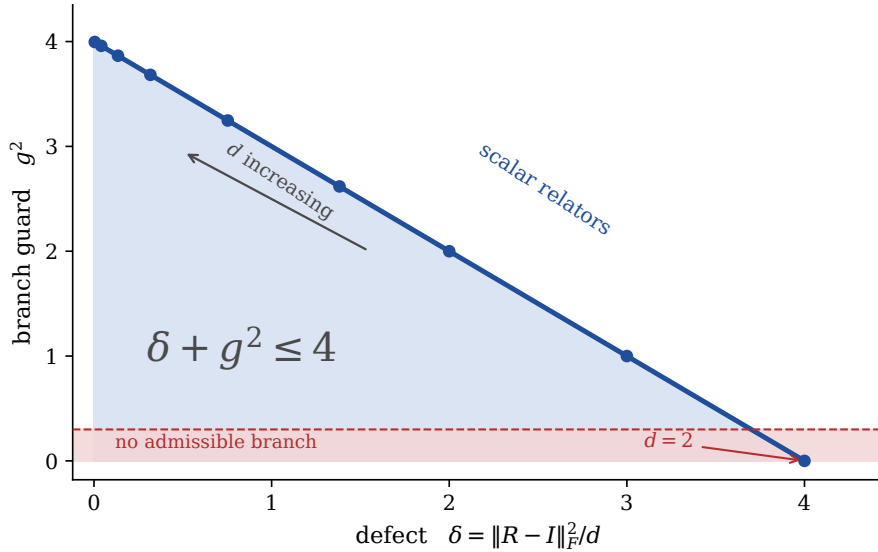


Figure 2: The admissible domain. Every anchor lies in the shaded triangle; scalar relators lie on its hypotenuse, where clock and shift sit at every d . Decreasing d slides the family into the corner at which the guard closes.

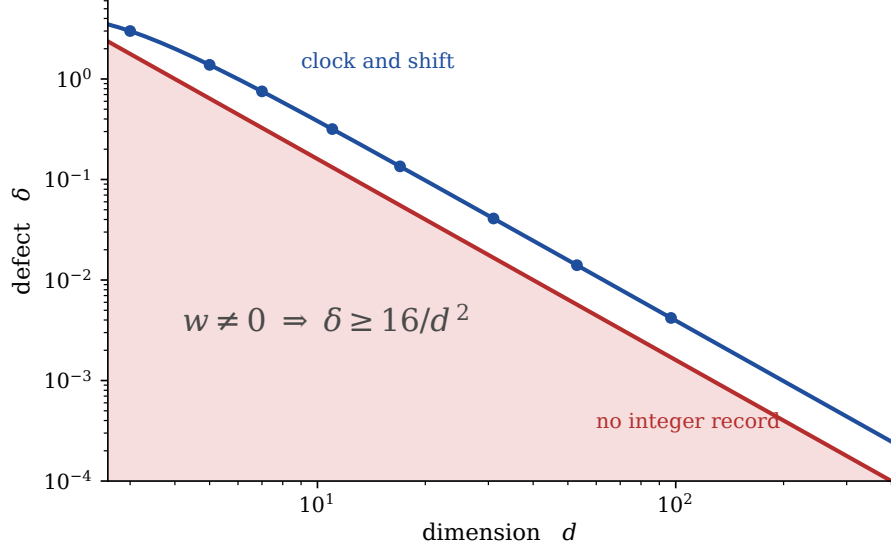


Figure 3: The amplitude floor at $w = 1$. Below the boundary no integer record is available at that dimension. Clock and shift ride a constant factor $\pi^2/4$ above the floor while carrying $w = 1$.

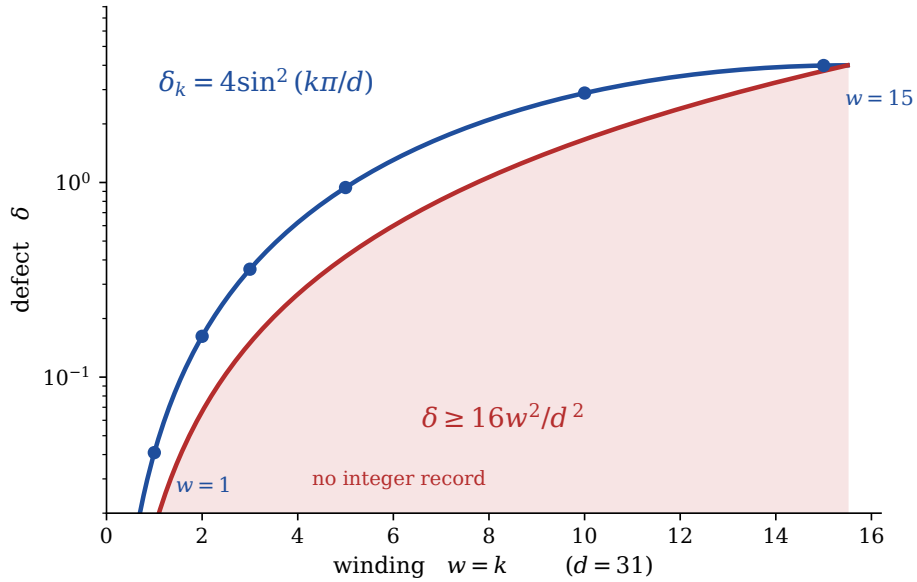


Figure 4: The winding ladder at $d = 31$. The quadratic floor $\delta \geq 16w^2/d^2$ and the realised family $\delta_k = 4 \sin^2(k\pi/d)$ converge as the winding approaches the branch edge: the gap closes from $\pi^2/4$ to 1, which is what makes the constant optimal.

4.3 The run, verbatim

```

=====
G1 INTEGRALITY AND THE SCALAR EQUALITY CASE: clock and shift.
   w = 1 at every d; delta + g^2 = 4 exactly; bound = (d/4)sqrt(delta) -> pi/2.
=====
d=3   exact          w=+1.000000000   w_chord=+1.000000000   delta=3.000000e+00   g=1.000000   delta+g^2=4.000000000   bound=1.2990 PASS
d=5   exact          w=+1.000000000   w_chord=+1.000000000   delta=1.381966e+00   g=1.618034   delta+g^2=4.000000000   bound=1.4695 PASS
d=7   exact          w=+1.000000000   w_chord=+1.000000000   delta=7.530204e-01   g=1.801938   delta+g^2=4.000000000   bound=1.5186 PASS
d=11  exact          w=+1.000000000   w_chord=+1.000000000   delta=3.174929e-01   g=1.918986   delta+g^2=4.000000000   bound=1.5495 PASS
d=17  exact          w=+1.000000000   w_chord=+1.000000000   delta=1.350555e-01   g=1.965946   delta+g^2=4.000000000   bound=1.5619 PASS
d=31  exact          w=+1.000000000   w_chord=+1.000000000   delta=4.094012e-02   g=1.989739   delta+g^2=4.000000000   bound=1.5681 PASS
d=53  exact          w=+1.000000000   w_chord=+1.000000000   delta=1.403781e-02   g=1.996487   delta+g^2=4.000000000   bound=1.5699 PASS
d=97  exact          w=+1.000000000   w_chord=+1.000000000   delta=4.194348e-03   g=1.998951   delta+g^2=4.000000000   bound=1.5705 PASS
=====

G1b THE WINDING LADDER: U -> U^k on clock and shift, d = 31.
   R = omega^k I so w = k and delta_k = 4 sin^2(k pi/d); the floor must clear k.
=====
k=1   w=+1.000000000   w_chord=+1.000000000   delta=0.040940117   pred=0.040940117   g=1.989739   bound=1.5681   (>= k)   delta/(16k^2/d^2)=2.458966 PASS
k=2   w=+2.000000000   w_chord=+2.000000000   delta=0.162084377   pred=0.162084377   g=1.959060   bound=3.1201   (>= k)   delta/(16k^2/d^2)=2.433798 PASS
k=3   w=+3.000000000   w_chord=+3.000000000   delta=0.358473118   pred=0.358473118   g=1.908279   bound=4.6401   (>= k)   delta/(16k^2/d^2)=2.392310 PASS
k=5   w=+5.000000000   w_chord=+5.000000000   delta=0.942071979   pred=0.942071979   g=1.748693   bound=7.5222   (>= k)   delta/(16k^2/d^2)=2.263328 PASS
k=10  w=+10.000000000   w_chord=+10.000000000   delta=2.880788303   pred=2.880788303   g=1.057928   bound=13.1540   (>= k)   delta/(16k^2/d^2)=1.730273 PASS
k=15  w=+15.000000000   w_chord=+15.000000000   delta=3.989738647   pred=3.989738647   g=0.101298   bound=15.4801   (>= k)   delta/(16k^2/d^2)=1.065039 PASS
pi^2/4 = 2.467401
=====

G2 NEGATIVE CONTROL: commuting pair. w = 0, delta = 0, g = 2.
=====
d=7   commuting      w=-0.000000000   w_chord=-0.000000000   delta=1.873837e-32   g=2.000000   delta+g^2=4.000000000   bound=0.0000 PASS
d=31  commuting      w=-0.000000000   w_chord=-0.000000000   delta=9.453904e-33   g=2.000000   delta+g^2=4.000000000   bound=0.0000 PASS
d=97  commuting      w=-0.000000000   w_chord=-0.000000000   delta=2.033120e-32   g=2.000000   delta+g^2=4.000000000   bound=0.0000 PASS
=====

G3 STABILITY AND THE STRICT INEQUALITY: kicked clock-shift, d=31.
   w stays 1; the relator leaves the scalar locus so delta + g^2 < 4.
=====
eta=1e-01   w=+1.000000000   w_chord=+1.000000000   delta=5.151479e-02   g=1.960698   delta+g^2=3.895850624   bound=1.7590 PASS
eta=1e-02   w=+1.000000000   w_chord=+1.000000000   delta=4.106643e-02   g=1.987454   delta+g^2=3.991039211   bound=1.5705 PASS
eta=1e-03   w=+1.000000000   w_chord=+1.000000000   delta=4.094131e-02   g=1.989513   delta+g^2=3.999103199   bound=1.5681 PASS
eta=1e-04   w=+1.000000000   w_chord=+1.000000000   delta=4.094013e-02   g=1.989719   delta+g^2=3.999921719   bound=1.5681 PASS
=====

G4 INVARIANCE: unitary conjugation and generator rephasing leave w fixed.
=====
baseline          w=+1.000000000   w_chord=+1.000000000   delta=4.094012e-02   g=1.989739   delta+g^2=4.000000000   bound=1.5681 PASS
conjugated         w=+1.000000000   w_chord=+1.000000000   delta=4.094012e-02   g=1.989739   delta+g^2=4.000000000   bound=1.5681 PASS
U -> e^{i th} U    w=+1.000000000   w_chord=+1.000000000   delta=4.094012e-02   g=1.989739   delta+g^2=4.000000000   bound=1.5681 PASS
both rephased      w=+1.000000000   w_chord=+1.000000000   delta=4.094012e-02   g=1.989739   delta+g^2=4.000000000   bound=1.5681 PASS
=====

G5 THE DOMAIN BOUND OVER RANDOM RELATORS: delta + g^2 <= 4, strict off
   the scalar locus. 1000 random unitary pairs, d = 12.
=====
violations: 0/1000   max delta+g^2 = 2.310779189   (bound 4)
=====

G5b BOUNDARY STRESS: the bound is approached, not sampled at random.
   Equality needs Re lambda_j constant, so drive the spectrum to a point.
=====
(a) near-commuting family: V rotated off a common eigenbasis by eta.
eta=1e-01   max delta+g^2 = 3.993267996   gap to 4 = 6.732e-03
eta=3e-02   max delta+g^2 = 3.999493685   gap to 4 = 5.063e-04
eta=1e-02   max delta+g^2 = 3.999941210   gap to 4 = 5.879e-05
eta=3e-03   max delta+g^2 = 3.999994121   gap to 4 = 5.879e-06
eta=1e-03   max delta+g^2 = 3.999999310   gap to 4 = 6.896e-07
(b) generic search for the equality locus: hill climb on delta + g^2,
d = 12, 6 restarts x 400 steps. Expected to fail; the locus is thin.
restart 1: delta+g^2 = 2.920020709   gap to 4 = 1.080e+00
restart 2: delta+g^2 = 3.050671820   gap to 4 = 9.493e-01
restart 3: delta+g^2 = 2.814474731   gap to 4 = 1.186e+00
restart 4: delta+g^2 = 2.840469780   gap to 4 = 1.160e+00
restart 5: delta+g^2 = 2.895491078   gap to 4 = 1.105e+00
restart 6: delta+g^2 = 2.914849274   gap to 4 = 1.085e+00
(c) along the hypotenuse: kicked clock-shift at eta=1e-3, several d.
d=3   delta^3.0000   max delta+g^2 = 3.998554432   gap to 4 = 1.446e-03
d=5   delta^1.3820   max delta+g^2 = 3.997870228   gap to 4 = 2.130e-03
d=7   delta^0.7530   max delta+g^2 = 3.997728220   gap to 4 = 2.272e-03
d=11  delta^0.3175   max delta+g^2 = 3.998442306   gap to 4 = 1.558e-03
d=17  delta^0.1351   max delta+g^2 = 3.998828265   gap to 4 = 1.172e-03
d=31  delta^0.0409   max delta+g^2 = 3.999286894   gap to 4 = 7.131e-04
violations: 0/1306   max delta+g^2 = 3.999999310   (bound 4)
=====

G6 THE BOUNDARY: d=2 places an eigenvalue at -1. The gate refuses.
=====
d=2   clock-shift      REFUSED g=1.22e-16 delta=4.000000   delta+g^2=4.000000000
d=4   clock-shift      w=+1.000000000   w_chord=+1.000000000   delta=2.000000e+00   g=1.414214   delta+g^2=4.000000000   bound=1.4142 PASS
=====

```

4.4 Reading the blocks

G1: the calibration returns $w = 1$ by both independent estimators at every d , with δ falling from 3.000 to 4.194×10^{-3} and $\delta + g^2 = 4.000000000$ on every row — the equality case, as the scalar relator requires.

G1b: the ladder returns $w = k$ exactly at $k = 1, 2, 3, 5, 10, 15$ in $d = 31$, with δ_k matching $4 \sin^2(k\pi/d)$ to twelve digits and the floor clearing k at every rung. The ratio $\delta_k/(16k^2/d^2)$ runs $2.458966 \rightarrow 1.065039$ against the predicted endpoints $\pi^2/4 = 2.467401$ and 1; the top rung $k = 15$ is $\lfloor (d-1)/2 \rfloor$, so the sharp integer bound permitted by the branch guard is attained on the same family that makes the floor’s constant optimal.

G2: commuting pairs give $w = 0$ and $\delta \sim 10^{-32}$.

G3: under kicks from $\eta = 10^{-1}$ to 10^{-4} the integer holds at 1 while the relator leaves the scalar locus, and $\delta + g^2$ drops strictly below 4 (3.895851 at the largest kick, rising to 3.999922 at the smallest) — the strict side of the equality clause, behaving as predicted.

G4: conjugation and either generator rephasing leave every digit unchanged.

G5: no violation of the domain bound in a thousand random unitary pairs at $d = 12$, largest observed 2.310779. This is a randomised sanity check over the interior, not a boundary test, and is recorded as one.

G5b: the boundary itself, approached deliberately rather than sampled. Driving the spectrum to a point through near-commuting pairs closes the gap to 4 as 6.7×10^{-3} , 5.1×10^{-4} , 5.9×10^{-5} , 5.9×10^{-6} , 6.9×10^{-7} ; walking the hypotenuse with kicked clock-shift at six dimensions holds the gap near 10^{-3} at defects from 3.0 down to 0.041, so the bound is approached from below all along the equality locus and not only in its corner. A generic search for the equality locus, hill-climbing $\delta + g^2$ from random starts, stalls at 3.05 — well short of what the constructed families reach. That failure is itself the finding: the equality locus is thin, and random optimisation does not locate an analytic manifold of measure zero. No violation in 1306 attempts.

G6: refusal at $d = 2$, with $g = 1.22 \times 10^{-16}$ and δ exactly 4 — the corner of the triangle.

5 The standing verdict

This route does not establish non-soficity, because the invariant it carries is stable in a stronger metric than the one soficity tests.

The Bott–Exel–Loring winding obstructs approximation in the operator norm. The testbed is the historical witness for exactly that: Voiculescu’s 1983 counterexample to the operator-norm question for almost-commuting unitaries is clock and shift [C]. Descend one rung and the obstruction is not merely weaker, it is provably absent: with respect to the normalised Hilbert–Schmidt norm, almost-commuting unitaries *are* near commuting ones, dimension-independently. Glebsky (2010) proves it for self-adjoint, unitary and normal matrices, and for several almost-commuting matrices at once; Filonov–Kachkovskiy (2010) give the self-adjoint case independently, Hadwin–Shulman (2018) place it inside the general tracial-stability framework, and Ioana (2024) supplies the negative counterpart for general matrices [C]. In the currency of this trace: on the normalised rung, $\mathbb{Z}^2$ is stable, so there is nothing there for a winding to obstruct.

That is the correction to the earlier phrasing that the reason was simply “the metric.” The reason is sharper. The winding currency is native to operator-norm and K -theoretic stability, where quantitative index separation is established; the transfer to normalised Hamming has no stability or separation theorem behind it, and one cannot be manufactured from the conversion $d_H = \frac{1}{2} \|\cdot\|_{\text{HS},n}^2$ alone. Arzhantseva and Păunescu make the same point from the other side: although the Hamming distance is expressible through the Hilbert–Schmidt distance, the Hilbert–Schmidt techniques for

unitaries do not deliver commutator stability for permutations, which is why that result had to be proved separately [C]. An exact conversion between two metrics does not carry theorems across them.

Rigidity on the wrong axis is blindness, not an asset [T].

What does the work over there, and it is not in this corpus. Kun’s theorem shows that the sofic approximation graphs of a group with property (T) are essentially vertex-disjoint unions of expanders. An expander has no sparse cut. Therefore the defect cannot be pushed into a thin seam and made to vanish by fraction. That anti-localisation step is the load-bearing piece of the whole construction [C]. Any programme aimed at “defect confinement” is re-deriving Kun 2016.

6 Two outside readings, assessed

Reading one restated the winding position correctly and raised two cautions, both of which were accepted. First, “same currency in two realisations” is the right conceptual compression, but the explicit normalisation identifying the smooth loop integral with w on a common family is not established, and was overstated; it is carried at [H] in the paper. Second, calling the boundary a rigidity wall is interpretation, not a consequence of the identity alone. Where the reading stopped short: it treated $\delta + g^2 = 4$ as a fact about clock and shift. It is the equality case of a general bound, and the general bound is the theorem — which is what converts the wall from interpretation into a proved domain statement.

Reading two proposed that the Hamming metric lets a topological defect escape to a vanishing boundary — a seam of size L in an area L^2 — and proposed an experiment to watch it happen. The instinct is right and the vintage is wrong. Its literature framing describes the state before 1 August. More to the point, the anti-seam mechanism it says is missing was proved in 2016: property (T) is exactly the condition preventing defect localisation, by Kun’s expander theorem. Of its three proposed outcomes, the third is the true one and it is already a theorem. The proposed experiment also hides its whole difficulty in one step — “translate the defect into a permutation action” — when the shift is a permutation matrix and the clock is not, and the impossibility of carrying that phase by permutations is the entire problem. The cited research note, read directly, is more careful than the use made of it: it states outright that no non-sofic group has been constructed there and isolates the remaining global statement as an open conjecture.

7 The chiral routes

7.1 The lead

If the medium tracing the currents is itself chiral — a Dirac operator, a chiral nematic, a non-Abelian vortex condensate — then the invariant it carries is not norm-valued and may survive the local surgery that Hamming forgives. The literature supports the class. Annala, Zamora-Zamora and Möttönen construct knots and links of non-Abelian vortices topologically protected against local reconnections and strand crossings, with the protection proved by an invariant of links coloured by the quaternion group Q_8 , realisable in dilute Bose–Einstein condensates [C]. The December 2025 chiral-nematic experiment tracks a conserved Hopf index through knot fusion and fission [C]. Helicity, being abelian, leaks through reconnections and is the wrong member of the family.

This is the correct class of invariant for a metric that forgives local cut-and-paste, and it is genuinely different from anything the gate measures. The two branches of the lead fail differently, and only one of them fails.

7.2 The Q_8 colouring branch: closed

A Q_8 colouring is a homomorphism from the complement's fundamental group into Q_8 up to conjugacy. Any invariant built by valuing or counting homomorphisms into a fixed finite group factors through the profinite completion, since a map to a finite group extends uniquely to it. That is why it survives surgery: it reads only the finite quotients, and local cut-and-paste does not change which finite quotients exist.

Now consider where soficity is in doubt. Residually finite implies sofic. LEF implies sofic. The load-bearing piece of the construction is a finitely presented infinite simple group — trivial profinite completion, no finite quotients at all — and the argument's punchline is that if the container were sofic then that piece would be LEF, which is impossible because finitely presented LEF groups are residually finite.

So on the only groups in play, every finite-group colouring invariant is identically trivial. Robustness, profiniteness and blindness are one property under three names [T]. The colouring survives every local move for the same reason it cannot see anything.

Confirmation from the other side: the invariant that does work in the approximation story is second bounded cohomology, and it is provably not a profinite invariant — there exist pairs of residually finite groups with isomorphic profinite completions, one with vanishing and one with non-vanishing real second bounded cohomology [C]. Non-profiniteness is the price of separating power.

7.3 The Hopf branch: open

The Hopf index is not a finite-group colouring. It is integer-valued, defined from a map to a target whose homotopy carries $\mathbb{Z}$, and it does not automatically factor through any profinite completion. The kill of the previous subsection has no purchase on it, and applying it there would be a category error.

What blocks this branch is different and weaker: no realisation of the index on the finite approximants has been constructed. Robustness under local surgery is free, but separation is not — the invariant must be computable on the approximating permutation systems and take a value the group cannot match, and nobody has built that map. The obstruction is

no finite-approximant realisation has been constructed,

not

profinite blindness.

The first is an absence of construction and can be repaired by construction. The second is a theorem and cannot [H for the assessment, C for the ingredients].

7.4 Why the two failures are different, and why it matters

Distinguishing them is not bookkeeping. A programme that treats both branches as killed by profiniteness will stop looking, and the thing it stops looking for — an integer-valued, surgery-stable invariant with a finite-approximant realisation — is precisely the shape of what the sofic metric would need. The Q_8 branch is closed by a theorem about what finite-group invariants can see. The Hopf branch is open and unattempted.

8 The two borrowed methods, taken seriously

8.1 Piccirillo, and the figure-eight slice

The trace move is not “swap in an equivalent object”. It is: determine what the invariant actually depends on, discover that the dependence is coarser than the object, then move to any representative where it is computable. Sliceness depends only on the 0-surgery trace, not the knot; find another knot with the same trace where the s -invariant is computable. All the leverage is in the coarseness of the dependence.

The figure-eight work in this corpus is the same move in a different category, and it is worth stating in those terms. The mixed Hessian at a singular point of the A -polynomial curve does not depend on the whole curve. Palindromicity in the longitude, $L^2 A(1/L, M) = A(L, M)$, means it depends only on behaviour along the fixed locus of the involution $L \mapsto 1/L$. On that locus,

$$A_{LM}(\pm 1, M) = \pm A_M(\pm 1, M)$$

exactly, so the mixed partial vanishes at every singular point lying there — all four nodes at once rather than the two the original argument reached [T].

What the slice revealed was two things. Negative: the apparent second-order degeneracy is not second order at all, because “ M is a repeated root of $A(1, M)$ ” and “ $A_M = 0$ ” are the same equation, so the phenomenon is an artifact of the amphichiral symmetry. Positive: the involution is the amphichirality; the tangent cones are axis-aligned in the canonical meridian–longitude coordinates for that reason; and at $(-1, \pm 1)$ the cones are definite, making those points acnodes.

Applied to the present problem, the same dependence analysis asks what soficity depends on. Not the presentation: soficity is closed in the space of marked groups, so non-soficity is open there (Ceccherini-Silberstein–Coornaert, Prop. 7.5.10) [C] — and openness is the trace-analogue, and it was cashed immediately — finite presentation came for free, while torsion-freeness required separate work precisely because torsion-freeness is not open. Any candidate invariant must be a function of the marked-group germ. That rules out a great deal before construction begins.

8.2 Viazovska, and where the certificate already is

The method has three parts, not one. Recognise the bound as linear programming over a positivity cone. Know the dual certificate’s exact sign conditions — the function and its Fourier transform each with prescribed signs beyond a radius. Construct it from a structure carrying the right symmetry, which is where modular forms enter in dimension eight.

The defect-floor statement one would want — nontrivial class implies defect fraction at least $c > 0$ — is a linear statement about the empirical distribution of disagreement, so a dual certificate exists in principle. But the sharper observation is that on one side the certificate already exists and it is the Kazhdan constant. For a group with property (T), every unitary representation without invariant vectors has some generator moving every unit vector by at least κ . That constant plays a role structurally analogous to the dual certificate’s: it forces a floor without inspecting any particular approximation, and it is why the expander theorem exists at all. Symmetry gives an explicit constant which forces the bound — the same architecture in both places [H for the identification, C for the ingredients]. It is an analogy between two mechanisms, not a claim that one is an instance of the other.

The frontier is therefore not “find a certificate”. It is: find one for groups without (T), where no symmetry hands you κ . That is well posed, and it is the one place the corpus has a native variable, since the $\sin^2 \theta \geq \varepsilon$ threshold is a Kazhdan-constant-shaped object — a uniform angular separation floor.

9 Kills ledger

What was killed	Why	Cost
The determinant winding (WIND-GATE-01)	det of a commutator is identically 1, so the endpoint carries nothing; and R^m never closes. Three of five blocks failed, one returning a non-integer.	One build, one run
“det is blind because $\zeta^d = 1$ ” as the root cause	True but not the cause. The cause is the path. The corrected statement is stronger.	One turn
Winding as a route to non-soficity	Operator-norm obstruction. On the normalised rung almost-commuting unitaries are near commuting ones by theorem, so there is nothing there to obstruct.	The session
“Normalised Hamming is normalised Hilbert–Schmidt”	Half its square, on permutation matrices. The factor was load-bearing for the metric argument.	One turn
“DGLT measures in the framework’s metric natively”	Their Frobenius is the <i>unnormalised</i> Hilbert–Schmidt norm. Ambient size does not drop out, which is the opposite of zero-and-span — and is why that rung has a counterexample and the one below it does not.	One turn
The defect-confinement experiment	The anti-localisation mechanism it proposed hunting for is Kun’s 2016 expander theorem. Also, its key step assumes phases can be carried by permutations, which is the problem itself.	One reading
Q_8 colouring invariants	They factor through the profinite completion. The groups in play have no finite quotients worth reading, so the invariants are identically trivial exactly where they are needed.	One turn
“The chiral route is closed”	Overreach. Only the Q_8 branch is closed. The Hopf index does not factor through finite quotients and stalls for a different and weaker reason.	One turn
“Piccirillo’s move is already taken”	Dismissive and wrong. The dependence analysis is the method, it is the same move as the figure-eight slice, and it yields a real constraint here.	Corrected in place
“Viazovska is only the shape of the problem”	Also wrong. The certificate is instantiated: it is the Kazhdan constant. The open part is groups without (T).	Corrected in place

10 What is left open

Non-MF, not non-sofic. The proposed winding route to non-soficity fell. Soficity itself fell independently and by other means; nothing in this trace contributed to that, and the gate established no non-soficity of its own. No priority is claimed here for the non-sofic existence result. MF and hyperlinear did not fall. Hyperlinear lives in normalised Hilbert–Schmidt and meets the same wall as this route. MF lives in the operator norm — the metric the winding currency actually reads. This is not a transfer: $\mathbb{Z}^2$ carries a nonzero Bott index and is MF regardless. But it is the first axis on which the metric agrees rather than disagrees [H].

A certificate without (T). A Kazhdan-constant analogue for a class of groups lacking property (T), expressed as a uniform floor in $\sin^2 \theta$. Hard, well posed, and the only one of the routes where the corpus supplies the natural variable [H].

A finite-approximant realisation of an integer chiral index. Section 7.3’s open branch. Not attempted here [H].

What went into the paper. The theorems of Section 4 are now a subsection of Module I in the projector geometry paper, with the figures and the gate in the appendix. That is the actual

yield, and it came out of a failed gate aimed somewhere else.

A The gate, as run

```
# SCRIPT: UPG-GATE-02b (Module I, noncommutative torus anchor: winding integrality
# and invariance; the admissible domain  $\delta + g^2 \leq 4$ ; the amplitude floor
#  $|w| \leq (d/4) \sqrt{\delta}$  and its higher-winding form  $\delta \geq 16 w^2 / d^2$ )
#
#  $R = U V U^\dagger V^\dagger$ ,  $\lambda_i = \text{eig}(R)$ 
#  $w = (1/2\pi) \sum_i \arg(\lambda_i)$  [principal branch]
#  $\delta = \|R - I\|_F^2 / d$ 
#  $g = \min_i |\lambda_i + 1|$  [branch guard]
# second estimator: winding of  $t \rightarrow \det((1-t)I + tR)$  on  $[0,1]$ 
#
# No spectral projection onto the unit circle is applied anywhere: the reported
# eigenvalues are those returned directly by NumPy for the computed relator.

import numpy as np

TAU = 1e-3
rng = np.random.default_rng(20260820)

def clock_shift(d):
    w = np.exp(2j*np.pi/d)
    U = np.diag([w**k for k in range(d)])
    V = np.roll(np.eye(d, dtype=complex), 1, axis=0)
    return U, V

def commuting_pair(d):
    U = np.diag(np.exp(1j*rng.uniform(0, 2*np.pi, d)))
    V = np.diag(np.exp(1j*rng.uniform(0, 2*np.pi, d)))
    return U, V

def relator(U, V):
    return U @ V @ U.conj().T @ V.conj().T

def unitary_kick(U, eta):
    d = U.shape[0]
    H = rng.normal(size=(d, d)) + 1j*rng.normal(size=(d, d))
    H = (H + H.conj().T)/2
    H /= np.linalg.norm(H, 2)
    ev, W = np.linalg.eigh(H)
    return U @ (W @ np.diag(np.exp(1j*eta*ev)) @ W.conj().T)

def readout(R):
    d = R.shape[0]
    lam = np.linalg.eigvals(R)
    g = np.min(np.abs(lam + 1.0))
    delta = np.linalg.norm(R - np.eye(d), 'fro')**2 / d
    w = np.sum(np.angle(lam)) / (2*np.pi)
    return lam, g, delta, w

def winding_chord(R, nt=20001):
    lam = np.linalg.eigvals(R)
    t = np.linspace(0.0, 1.0, nt)[: , None]
    ang = np.unwrap(np.angle((1.0 - t) + t*lam[None, :]), axis=0)
    return np.sum(ang[-1] - ang[0]) / (2*np.pi)

def report(tag, R, expect=None):
    d = R.shape[0]
    lam, g, delta, w = readout(R)
```

```

if g < TAU:
    print(f"{tag:<32s} REFUSED g={g:.2e} delta={delta:.6f}"
          f" delta+g^2={delta+g*g:.9f}")
    return
w2 = winding_chord(R)
floor = d*np.sqrt(delta)/4
ok = "" if expect is None else (
    " PASS" if abs(w-expect) < 1e-9 and abs(w2-expect) < 1e-6 else " FAIL")
print(f"{tag:<32s} w={w:.9f} w_chord={w2:.9f} delta={delta:.6e}"
      f" g={g:.6f} delta+g^2={delta+g*g:.9f} bound={floor:.4f}{ok}")

print("="*116)
print("G1 INTEGRALITY AND THE SCALAR EQUALITY CASE: clock and shift.")
print("    w = 1 at every d; delta + g^2 = 4 exactly; bound = (d/4)sqrt(delta) -> pi/2.")
print("="*116)
for d in [3, 5, 7, 11, 17, 31, 53, 97]:
    U, V = clock_shift(d)
    report(f"    d={d:<4d} exact", relator(U, V), expect=1.0)
print()

print("="*116)
print("G1b THE WINDING LADDER: U -> U^k on clock and shift, d = 31.")
print("    R = omega^k I so w = k and delta_k = 4 sin^2(k pi/d); the floor must clear k.")
print("="*116)
d = 31
U0, V0 = clock_shift(d)
for k in [1, 2, 3, 5, 10, 15]:
    R = relator(np.linalg.matrix_power(U0, k), V0)
    _, g, delta, w = readout(R)
    w2 = winding_chord(R)
    pred = 4*np.sin(k*np.pi/d)**2
    floor = d*np.sqrt(delta)/4
    ratio = delta / (16*k*k/(d*d))
    ok = " PASS" if (abs(w-k) < 1e-9 and abs(w2-k) < 1e-6
                    and abs(delta-pred) < 1e-12 and floor >= k - 1e-12) else " FAIL"
    print(f"    k={k:<3d} w={w:.9f} w_chord={w2:.9f} delta={delta:.9f}"
          f"    pred={pred:.9f} g={g:.6f} bound={floor:.4f} (>= k)"
          f"    delta/(16k^2/d^2)={ratio:.6f}{ok}")
print(f"    pi^2/4 = {np.pi**2/4:.6f}")
print()

print("="*116)
print("G2 NEGATIVE CONTROL: commuting pair. w = 0, delta = 0, g = 2.")
print("="*116)
for d in [7, 31, 97]:
    U, V = commuting_pair(d)
    report(f"    d={d:<4d} commuting", relator(U, V), expect=0.0)
print()

print("="*116)
print("G3 STABILITY AND THE STRICT INEQUALITY: kicked clock-shift, d=31.")
print("    w stays 1; the relator leaves the scalar locus so delta + g^2 < 4.")
print("="*116)
for eta in [1e-1, 1e-2, 1e-3, 1e-4]:
    U, V = clock_shift(31)
    report(f"    eta={eta:.0e}", relator(unitary_kick(U, eta),
                                         unitary_kick(V, eta)), expect=1.0)
print()

```

```

print("="*116)
print("G4 INVARIANCE: unitary conjugation and generator rephasing leave w fixed.")
print("="*116)
U, V = clock_shift(31)
report(" baseline", relator(U, V), expect=1.0)
X = rng.normal(size=(31, 31)) + 1j*rng.normal(size=(31, 31))
Q, _ = np.linalg.qr(X)
report(" conjugated", relator(Q @ U @ Q.conj().T, Q @ V @ Q.conj().T), expect=1.0)
report(" U -> e^{i th} U", relator(np.exp(0.7371j)*U, V), expect=1.0)
report(" both rephased", relator(np.exp(0.7371j)*U, np.exp(-2.11j)*V), expect=1.0)
print()

print("="*116)
print("G5 THE DOMAIN BOUND OVER RANDOM RELATORS: delta + g^2 <= 4, strict off")
print(" the scalar locus. 1000 random unitary pairs, d = 12.")
print("="*116)
worst = -1.0
viol = 0
for _ in range(1000):
    A = rng.normal(size=(12, 12)) + 1j*rng.normal(size=(12, 12))
    B = rng.normal(size=(12, 12)) + 1j*rng.normal(size=(12, 12))
    QA, _ = np.linalg.qr(A)
    QB, _ = np.linalg.qr(B)
    _, g, delta, _ = readout(relator(QA, QB))
    s = delta + g*g
    worst = max(worst, s)
    if s > 4 + 1e-9:
        viol += 1
print(f" violations: {viol}/1000 max delta+g^2 = {worst:.9f} (bound 4)")
print()

print("="*116)
print("G5b BOUNDARY STRESS: the bound is approached, not sampled at random.")
print(" Equality needs Re lambda_j constant, so drive the spectrum to a point.")
print("="*116)

def haar(d):
    X = rng.normal(size=(d, d)) + 1j*rng.normal(size=(d, d))
    Q, Rr = np.linalg.qr(X)
    return Q @ np.diag(np.diag(Rr)/np.abs(np.diag(Rr)))

d = 12
print(" (a) near-commuting family: V rotated off a common eigenbasis by eta.")
grand = -1.0
grandviol = 0
for eta in [1e-1, 3e-2, 1e-2, 3e-3, 1e-3]:
    worst_e = -1.0
    for _ in range(200):
        W = haar(d)
        U = W @ np.diag(np.exp(1j*rng.uniform(0, 2*np.pi, d))) @ W.conj().T
        V = W @ np.diag(np.exp(1j*rng.uniform(0, 2*np.pi, d))) @ W.conj().T
        _, g, delta, _ = readout(relator(U, unitary_kick(V, eta)))
        s = delta + g*g
        worst_e = max(worst_e, s)
        if s > 4 + 1e-9:
            grandviol += 1
    grand = max(grand, worst_e)
print(f" eta={eta:.0e} max delta+g^2 = {worst_e:.9f} gap to 4 = {4-worst_e:.3e}")

```

```

print(" (b) generic search for the equality locus: hill climb on  $\delta + g^2$ ,")
print("      d = 12, 6 restarts x 400 steps. Expected to fail; the locus is thin.")
def score(U, V):
    _, g, delta, _ = readout(relator(U, V))
    return delta + g*g
best = -1.0
for restart in range(6):
    U, V = haar(d), haar(d)
    s = score(U, V)
    step = 0.3
    for it in range(400):
        Un, Vn = unitary_kick(U, step), unitary_kick(V, step)
        sn = score(Un, Vn)
        if sn > s:
            U, V, s = Un, Vn, sn
        else:
            step *= 0.995
    if s > 4 + 1e-9:
        grandviol += 1
    best = max(best, s)
    print(f"      restart {restart+1}:  $\delta + g^2 = \{s:.9f\}$     gap to 4 =  $\{4-s:.3e\}$ ")
grand = max(grand, best)
print(" (c) along the hypotenuse: kicked clock-shift at  $\eta=1e-3$ , several d.")
for dd in [3, 5, 7, 11, 17, 31]:
    Uc, Vc = clock_shift(dd)
    worst_d = -1.0
    for _ in range(50):
        _, g, delta, _ = readout(relator(unitary_kick(Uc, 1e-3),
                                           unitary_kick(Vc, 1e-3)))

        s = delta + g*g
        worst_d = max(worst_d, s)
        if s > 4 + 1e-9:
            grandviol += 1
    grand = max(grand, worst_d)
    print(f"      d={dd:<3d}  $\delta + g^2 = \{4 * \sin(\pi/dd)^2:.4f\}$ "
          f"      max  $\delta + g^2 = \{worst_d:.9f\}$     gap to 4 =  $\{4-worst_d:.3e\}$ ")
print(f" violations: {grandviol}/1306 max  $\delta + g^2 = \{grand:.9f\}$  (bound 4)")
print()

print("="*116)
print("G6 THE BOUNDARY: d=2 places an eigenvalue at -1. The gate refuses.")
print("="*116)
U, V = clock_shift(2)
report(" d=2 clock-shift", relator(U, V))
U, V = clock_shift(4)
report(" d=4 clock-shift", relator(U, V), expect=1.0)

```